1	Blockchain Technology: Cryptocurrency and Applications	S. Shukla, M. Dhawan, S. Sharma, S. Venkatesan	Oxford University Press	2019
2	Bitcoin and cryptocurrency technologies: a comprehensive introduction	Arvind Narayanan et. Al	Princeton University Press	2016
Reference Book(s):				
1	Research perspectives and challenges for Bitcoin and cryptocurrency	Joseph Bonneau et al	SoK, IEEE Symposium on security and Privacy	2015
2	The bitcoin backbone protocol - analysis and applications	J.A.Garay et al	EUROCRYPT LNCS VOl 9057, (VOLII), pp	2015
3	Analysis of Blockchain protocol in Asynchronous networks	R.Pass et al	EUROCRYPT	2017
4	Fruitchain, a fair blockchain	R.Pass et al	PODC	2017
5	Blockchain: The Blockchain for Beginnings, Guild to Blockchain Technology and Blockchain Programming	Josh Thompson	Create Space Independent Publishing Platform	2017

(Group-6): Course Code :20SFC252 : Data Base Security	
Exam Hours: 3 hours	**Exam Marks(Maximum):100**
Module-1	
Introduction: Introduction to Databases, Security Problems in Databases Security Controls Conclusions. Security Models 1: Introduction, Access Matrix Model, Take-Grant Model, Acten Model, PN Model, Hartson and Hsiao's Model, Fernandez's Model, Bussolati and Martella's Model for Distributed databases.	
Module-2	
Security Models 2: Bell and LaPadula's Model, Biba's Model, Dion's Model, Sea View Model, Jajodia and Sandhu's Model, The Lattice Model for the Flow Control conclusion. Security Mechanisms: Introduction, User Identification/Authentication, Memory Protection, Resource Protection, Control Flow Mechanisms, Isolation, Security Functionalities in Some Operating Systems, Trusted Computer System, Evaluation Criteria.	
Module-3	
Security Software Design: Introduction, A Methodological Approach to Security, Software Design, Secure Operating System Design, Secure DBMS Design, Security Packages, Database Security Design.	
Module-4	

Statistical Database Protection & Intrusion Detection Systems: Introduction, Statistics, Concepts and Definitions, Types of Attacks, Inference Controls, evaluation Criteria for Control Comparison, Introduction IDES System, RETISS System, ASES System Discovery.

Module-5

Models For The Protection Of New Generation Database Systems 1: Introduction, A Model for the Protection of Frame Based Systems, A Model for the Protection of Object-Oriented Systems, SORION Model for the Protection of Object-Oriented Databases. Models For The Protection Of New Generation Database Systems 2: A Model for the Protection of New Generation Database Systems, the Orion Model, Jajodia and Kogan's Model, A Model for the Protection of Active Databases Conclusions.

Question paper pattern:

- The question paper will have ten questions.
- Each full question is for 20 marks.
- There will be 2full questions (with a maximum of four sub questions in one full question) from each module.
- Each full question with sub questions will cover the contents under a module.
- Students will have to answer 5 full questions, selecting one full question from each module.■

Textbook/Reference Books

	Title of the book	Author Name	Publisher's Name	Publication year
Textbook(s):				
1	Database Security and Auditing	Hassan A. Afyoun	CENGAGE Learning	2009
2	Database Security	Castano	Pearson Education	1995
Reference Book(s):				
1	Database security	Alfred Basta, Melissa Zgola	CENGAGE learning	2014

(Group-6): Course Code :20SCE253
Course Title : Decision Support System

Exam Hours: 3 hours	**Exam Marks(Maximum):100**

Module-1

Introduction to decision support systems: DSS Defined, History of decision support systems, Ingredients of a DSS, Data and model management, DSS Knowledge base, User interfaces, User interfaces, The DSS user, Categories and classes of DSSs, Chapter Summary. Decisions and decision makers Decision makers: who are they, Decision styles, Decision effectiveness, How can a DSS help?, A Typology of decisions, Decision theory and simon's model of problem solving, Bounded decision making, The process of choice, Cognitive processes, Biases and heuristics in decision making, Chapter summary.

Module-2

Decisions in the organization: Understanding the organization, Organizational culture. Modelling decision processes: Defining the problem and its structures, Decision models, Types of probability, Techniques for forecasting probabilities, Calibration and sensitivity, Chapter summary

Module-3